calm_weighted_filter

equity strategy | family: equity-volatility | status: research | derived from `balanced_leverage`

What this strategy does

Calm Weighted plus the Volatility Filter

Why it should work. Skip the cells that history says do not pay -- but the data shows a gradient, not a bad cell.

Design notes (from the strategy's source file).

Volatility Filter (and Calm Weighted Filter) -- skip stock groups that have not paid, judged only on past data.

Drops a (side, idio-vol tercile) cell from new entries when its own history says it has not earned its keep, judged ONLY on quarters whose positions had fully exited. Result: filtering does not help on its own (corrected run: Sharpe 0.89 vs 1.09 for Balanced Leverage; pre-fix 1.23 vs 1.30; the effect is a smooth gradient, not one broken cell) and adds nothing once inverse-vol sizing is on (corrected: Calm Weighted Filter 1.08 < Calm Weighted 1.24; pre-fix 1.41 < 1.47). Kept as a documented negative result: size by vol, do not filter by it.

parameter	value
base_unit_fraction	0.003
max_gross_leverage	2.5
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	priority
exposure_cap	open_book
beta_overlay	0.0

Performance

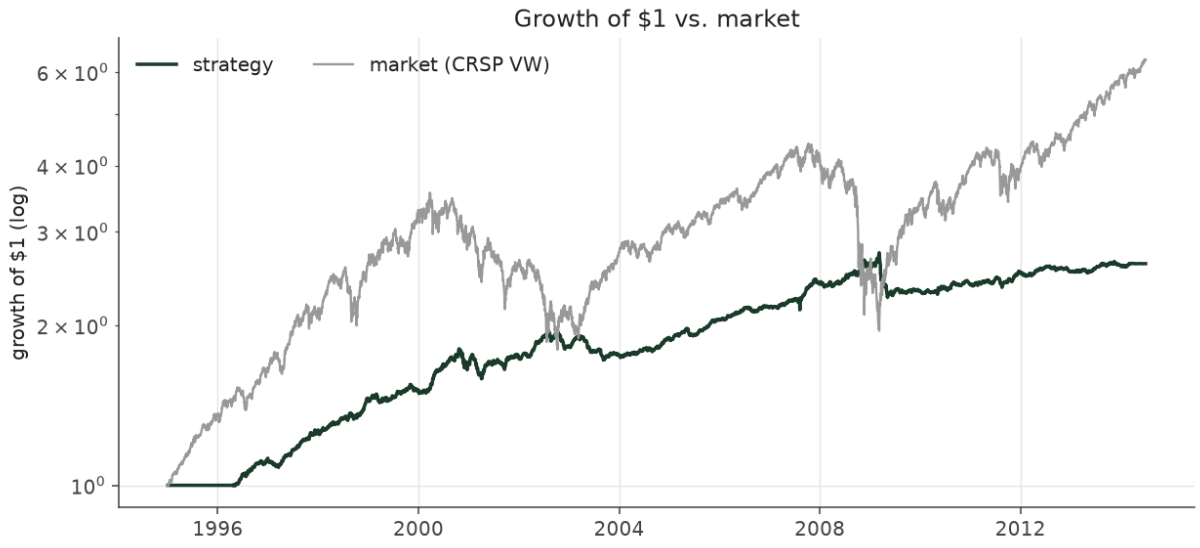
metric	calm weighted filter	market (CRSP VW)
Annualized return	5.06%	9.94%
Annualized volatility	4.66%	19.35%
Sharpe (NAV-pct)	1.08	0.59
Sortino	1.56	
Max drawdown	-17.6%	-55.5%
Calmar	0.29	
Final NAV ($10M start)	$26,144,178	
Worst day	-3.94%	
% positive days	50.1%	
Sharpe 1996-2001	1.79	
Sharpe 2002-2007	1.22	
Sharpe 2008-2013	0.30	

Correlation of daily returns with the market: **-0.01**.

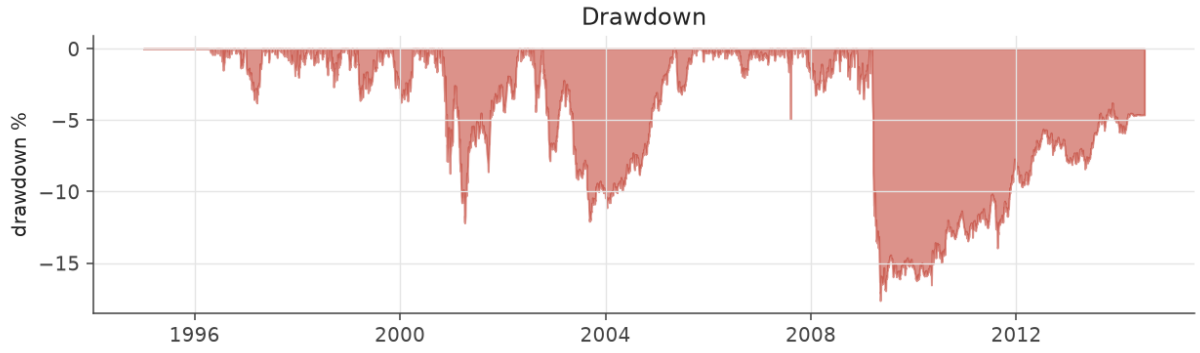
• best Sharpe among equity: `calm_rank_tilt` (1.47); this one: 1.08

• best return among equity: `market_boost_max` (0.116); this one: 0.0506

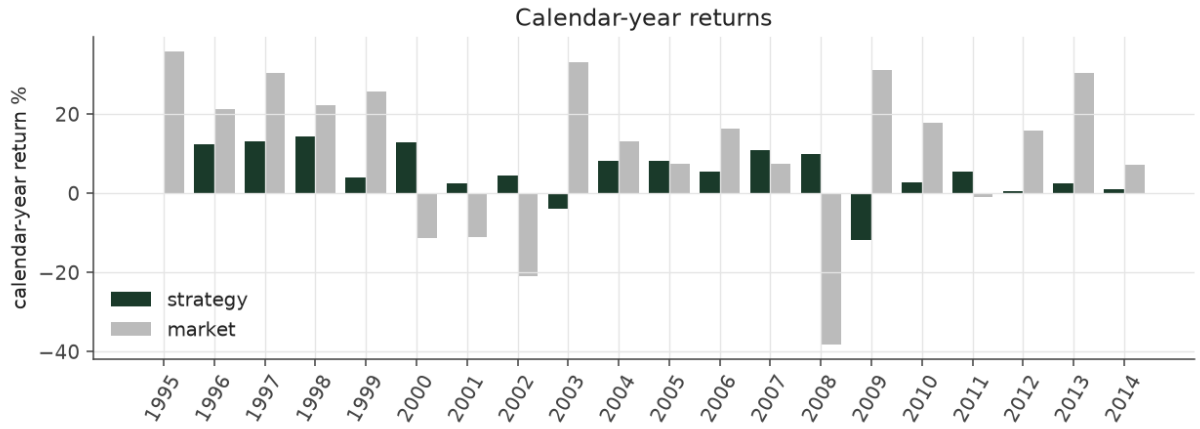
• best Calmar among equity: `calm_rank_tilt` (0.442); this one: 0.287



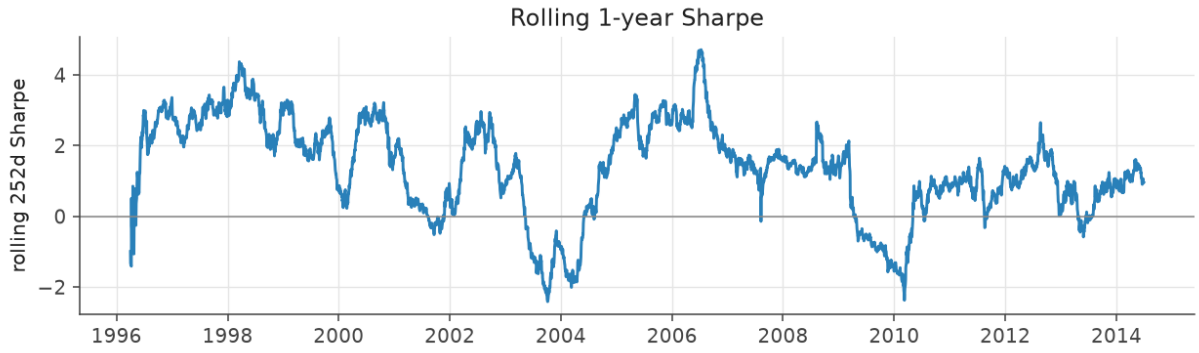
NAV (log scale)



Drawdown from running peak

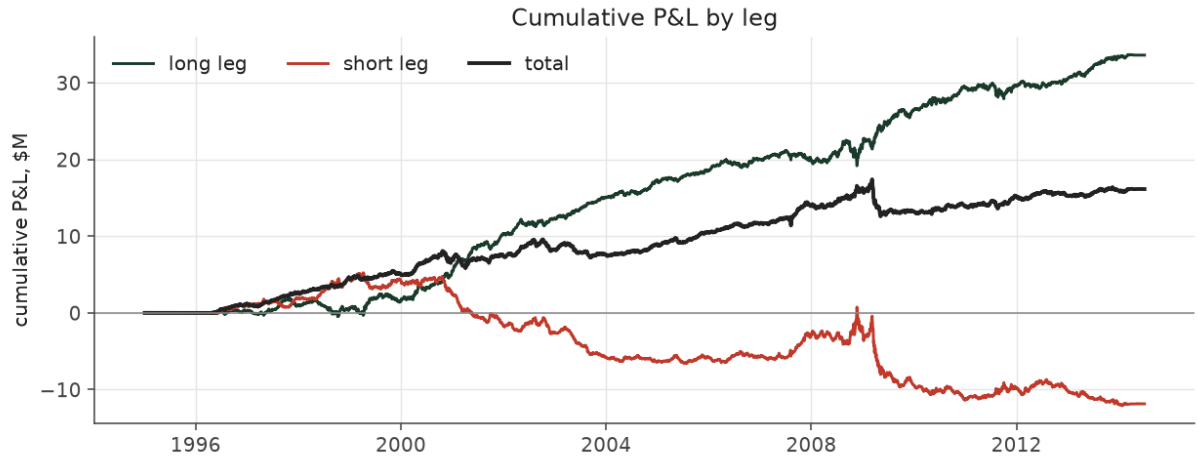


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	33.61	-11.85
P&L after costs ($M)	30.87	-14.73
of which common (average-event) component ($M)	21.76	-22.12
of which excess / PEAD component ($M)	11.84	10.26
excess minus costs ($M)	9.11	7.39
gross return per $ traded (bp)	247.89	-86.16
positions in the money (hit rate)	0.56	0.48
stand-alone Sharpe (leg only)	1.27	-0.35

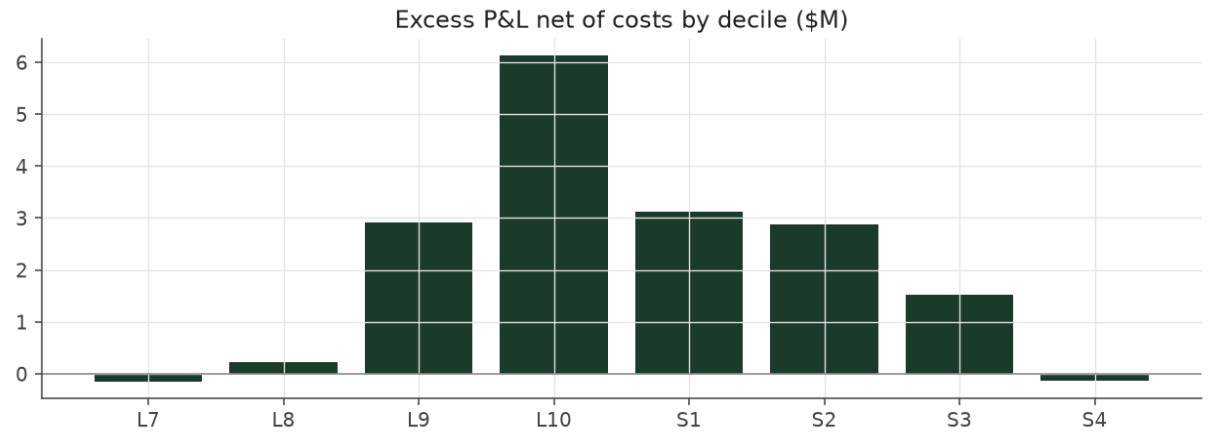
Daily P&L correlation between the two legs: **-0.78**.

• **The short leg loses money in absolute terms** (-11.9M before costs) while the long leg earns +33.6M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 21.8M / 22.1M. Stripping that common component, the surprise signal itself earned +11.8M on the long side and +10.3M on the short side.

• The two legs' daily P&L are -0.78 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 1.27 long / -0.35 short vs 1.08 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	7	5,871	89.22	1.37	0.03	0.18	-0.16	-0.9%	-17.51
long	8	15,155	299.11	5.64	0.84	0.60	0.23	1.4%	7.77
long	9	17,504	416.74	8.68	3.74	0.83	2.90	17.6%	69.70
long	10	19,169	550.70	15.18	7.24	1.11	6.13	37.2%	111.32
short	1	17,106	542.00	-5.99	4.30	1.18	3.12	18.9%	57.61
short	2	17,545	473.90	-4.81	3.85	0.98	2.88	17.4%	60.69
short	3	13,895	320.77	-3.27	2.18	0.65	1.53	9.3%	47.64
short	4	2,188	39.07	-0.66	-0.07	0.07	-0.14	-0.8%	-35.64



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	13,004	255.30	0.39	1.95	0.99	0.95	5.8%	37.37
2	20,713	514.35	5.21	6.96	1.47	5.49	33.3%	106.67
3	23,276	621.98	4.67	5.96	1.33	4.62	28.0%	74.31
4	25,115	665.12	3.69	4.51	1.07	3.44	20.9%	51.75
5	26,325	674.75	2.18	2.74	0.75	1.99	12.1%	29.55

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	12,449	196.55	2.26	3.60	0.98	2.62	15.9%	133.42
2	21,246	575.59	3.51	6.05	1.73	4.32	26.2%	75.11
3	23,678	680.72	5.17	6.64	1.36	5.27	32.0%	77.48
4	25,037	650.42	3.36	3.77	0.91	2.85	17.3%	43.88

5	26,019	627.99	1.80	2.00	0.63	1.37	8.3%	21.82

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	38,507	942.57	0.42	7.51	1.82	5.70	34.5%	60.41
41-60d	25,725	689.22	2.88	5.70	1.54	4.17	25.3%	60.47
61-90d	35,271	885.01	7.93	4.68	1.65	3.03	18.4%	34.21
>90d	8,930	214.70	4.91	4.21	0.60	3.61	21.9%	168.08

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	31,425	613.57	7.59	8.76	1.23	7.53	45.7%	122.77
2002-07	34,959	926.36	6.15	7.96	1.86	6.10	37.0%	65.84
2008-13	42,049	1,191.57	2.41	5.39	2.52	2.87	17.4%	24.06

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	18,852	261.58	0.75	1.25	0.57	0.69	4.2%	26.36
Chems	2,525	185.06	0.65	1.00	0.33	0.67	4.0%	35.97
Durbl	2,773	200.88	1.43	1.90	0.38	1.52	9.2%	75.77
Enrgy	6,141	225.35	2.43	2.88	0.43	2.44	14.8%	108.35
Hlth	10,512	259.75	0.94	1.48	0.55	0.92	5.6%	35.56
Manuf	11,360	244.18	1.24	1.86	0.52	1.34	8.1%	54.91
Money	15,637	228.86	2.01	2.59	0.53	2.06	12.5%	89.94
NoDur	4,847	224.64	0.95	1.36	0.44	0.92	5.6%	40.87
Other	16,250	250.10	1.98	2.60	0.57	2.03	12.3%	81.05
Shops	11,154	254.68	2.43	2.95	0.54	2.41	14.6%	94.58
Telcm	3,734	220.55	1.21	1.61	0.41	1.20	7.3%	54.42
Utils	4,648	175.88	0.11	0.63	0.33	0.30	1.8%	17.22

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	33,394	606.86	-0.11	1.82	1.14	0.69	4.2%	11.31
low vol	37,832	1,166.70	8.85	11.57	2.56	9.01	54.6%	77.24
mid vol	35,470	912.90	6.54	7.77	1.82	5.95	36.1%	65.22

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	5.06%	-0.00%
0.5%	1.4	4.77%	0.29%
1.0%	2.8	4.46%	0.60%
2.0%	5.5	3.78%	1.28%
5.0%	13.8	1.07%	3.99%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.